

The problem is BLIND: one measurement, two unknowns

P cannot be corrected without a clean x — and x cannot be reconstructed without the right P

$$y = A_{P(\theta)} x$$

$P(\theta) = P_{nom} T(\theta)$ — the nominal geometry,
times the rigid motion we did not see

y — **KNOWN**

the measured projections
 $360 \times 500 \times 700 = 126$ M values

θ — **UNKNOWN**

6 DoF per view (a rigid head pose)
 $6 \times 360 = 2\,160$ parameters

x — **UNKNOWN**

the volume we are after
 $256^3 = 16.8$ M voxels

Correcting P needs a clean x

$$\hat{\theta} = \arg \min_{\theta} \|A_{P(\theta)} x - y\|^2$$

the fit is only as good as the image you hand it —
a streaked x happily explains the streaks.

MEASURED (our estimator sweep): from the COLD FDK
every configuration plateaus at $\approx 2.0^\circ$; from a clean
reference the same budget reaches $0.09\text{--}0.27^\circ$.

**CIRCULAR
DEPENDENCY**

Reconstructing x needs the true P

$$\hat{x} = \arg \min_x \|A_{P(\theta)} x - y\|^2$$

a wrong θ does not just blur — it BAKES its own
artefacts into whatever image the solver returns.

MEASURED (same loop, oracle θ from step 0):
 x_t reaches 40.45 dB with the true geometry,
38.16 dB when θ has to be estimated blind.

hand it a better $\hat{\theta}$

Counting says it is solvable — the difficulty is elsewhere

126 M measurements vs $16.8\text{ M} + 2\,160$ unknowns: NOT underdetermined.

- the coupling is BILINEAR in (x, θ) — the joint problem is non-convex, and every local minimum is an image that explains the data
- an EXACT SE(3) GAUGE: (x, θ) and $(g \cdot x, \theta \circ g^{-1})$ give the SAME y , so the solution is an orbit, never a single pose
- per-view translation ALONG THE BEAM is nearly unobservable (92% of a converged estimator's residual lives there)

What breaks the cycle

$$\min_{x, \theta} \frac{1}{2} \|A_{P(\theta)} x - y\|^2 + \lambda \text{TV}(x) \quad + \quad \text{a learned prior on } x$$

1. the PRIOR supplies what the data cannot: it improves x without knowing θ
2. ALTERNATE (Gauss–Seidel): fit θ on the improved x , then solve x under it
3. the GEOMETRY BRIDGE keeps every intermediate x a real reconstruction, so the θ fit never runs on an image the operator cannot produce
4. score modulo the gauge: rigid-align before PSNR/SSIM, report RPE for θ